

DETROIT CITY, MI, USA

WMO: 725375

Lat: 42.409N

Lon: 83.010W

Elev: 626

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	8.8	-6.1	4.1	5.7	-1.5	5.2	10.5	24.3	28.1	21.6	30.0	9.2	250	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	90.5	73.2	87.9	71.9	85.3	70.5	76.2	85.7	74.5	83.6	73.0	81.8	11.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	126.5	81.2	71.7	119.9	79.9	70.0	113.1	78.6	40.0	85.9	38.4	83.6	37.0	81.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.6	18.6	16.9	-1.5	95.2	6.4	3.2	-6.2	97.5	-9.9	99.4	-13.6	101.2	-18.2	103.5		
			-2.5	79.5	6.3	2.1	-7.0	81.0	-10.7	82.2	-14.2	83.4	-18.7	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	26.2	28.5	36.9	48.7	60.0	70.2	74.5	72.9	65.9	53.8	41.8	31.9
	DBStd	19.02	10.72	10.11	10.48	8.83	8.43	6.93	5.40	5.16	7.20	8.27	8.73	9.19
	HDD50	2799	739	604	427	126	11	0	0	0	1	52	274	565
	HDD65	5988	1203	1023	871	494	205	27	2	5	76	359	697	1026
	CDD50	3186	1	2	22	86	320	607	760	710	477	171	27	3
	CDD65	901	0	0	1	4	49	185	297	249	103	13	0	0
	CDH74	7413	0	0	10	55	465	1602	2614	1866	726	75	0	0
	CDH80	2284	0	0	1	7	127	547	883	517	191	11	0	0
Wind	WSAvg	8.4	9.7	9.4	9.3	9.6	8.3	7.5	7.1	6.6	6.9	8.1	8.8	9.1
Precipitation	PrecAvg	33.9	2.4	2.2	2.4	3.0	3.7	3.3	3.2	2.8	3.0	2.7	2.6	2.5
	PrecMax	44.2	4.1	5.2	4.2	4.8	7.8	5.6	6.2	5.9	6.1	6.7	4.4	4.4
	PrecMin	27.4	0.8	0.6	0.8	0.6	1.5	1.6	1.3	0.5	0.8	0.4	0.6	0.8
	PrecStd	4.3	1.0	1.2	1.0	1.0	1.6	1.2	1.3	1.4	1.7	1.4	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	60.9	73.3	80.7	87.9	92.8	94.6	91.9	89.7	81.1	67.9	59.3
		MCWB	52.9	53.7	60.3	62.9	69.3	73.7	76.2	74.9	72.9	68.1	57.3	55.1
	2%	DB	51.5	52.7	65.0	74.1	83.7	89.8	90.7	88.2	85.5	75.4	62.6	53.9
		MCWB	48.4	47.4	54.8	58.9	67.3	72.1	73.5	72.9	70.1	64.4	55.6	50.2
	5%	DB	45.5	46.5	59.0	69.0	79.5	86.2	87.9	85.4	81.5	71.5	58.9	48.4
		MCWB	41.7	41.2	50.5	56.1	65.5	70.4	72.2	71.4	68.4	62.2	52.8	44.3
	10%	DB	40.1	41.3	53.0	63.8	74.9	82.8	84.9	82.6	78.2	67.4	55.0	43.7
		MCWB	37.0	36.9	45.9	53.0	63.1	68.6	70.7	70.0	66.7	58.9	49.3	39.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	55.1	61.8	65.6	72.6	76.3	79.2	78.3	75.6	70.1	60.4	55.6
		MCDB	55.8	58.3	72.6	76.9	83.9	89.1	90.2	87.9	85.5	78.3	64.3	58.3
	2%	WB	48.4	47.8	56.6	61.6	69.8	74.2	76.5	75.8	73.1	67.1	57.1	50.5
		MCDB	50.5	51.4	62.8	69.8	79.6	85.8	86.1	83.7	81.2	73.2	61.3	53.0
	5%	WB	41.5	41.2	51.8	57.8	67.6	72.6	74.8	73.9	71.0	63.4	53.5	44.7
		MCDB	44.8	46.3	57.9	66.6	76.6	83.1	83.6	81.3	78.0	69.3	57.9	48.0
	10%	WB	37.0	37.3	45.9	54.1	64.7	70.8	73.1	72.2	68.9	60.2	49.9	40.2
		MCDB	39.7	41.3	52.3	62.9	73.6	79.9	81.7	79.8	75.5	66.1	54.5	43.4
Mean Daily Temperature Range	5% DB	MDBR	12.0	13.0	15.5	18.1	18.4	17.7	17.2	16.2	17.0	15.4	12.7	10.5
		MCDBR	16.7	19.2	23.5	25.5	22.6	20.5	19.2	18.2	20.3	20.4	17.7	15.5
		MCWBR	15.0	15.5	15.9	15.5	12.3	9.7	8.5	8.3	9.3	11.5	13.6	13.2
	5% WB	MCDBR	16.3	18.7	22.3	23.0	20.3	18.6	17.4	16.0	17.1	17.8	16.6	14.9
		MCWBR	15.3	15.8	16.7	16.2	12.9	9.8	9.3	8.8	9.2	11.5	14.1	13.8
Clear-Sky Solar Irradiance	taub	0.302	0.334	0.352	0.398	0.429	0.446	0.436	0.428	0.389	0.360	0.320	0.309	
	taud	2.395	2.287	2.353	2.247	2.209	2.201	2.253	2.265	2.355	2.418	2.494	2.420	
	Ebn at Noon	266	272	282	276	269	264	265	264	266	261	259	252	
	Edh at Noon	25	33	35	42	45	45	43	41	35	29	23	23	
All-Sky Solar Radiation	RadAvg	465	744	1113	1467	1715	1938	1935	1677	1353	880	545	385	
	RadStd	49	85	112	128	147	121	96	67	117	75	70	31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	N/A

Nomenclature: See separate page